

A PROJECT TITLE
ON
customer segmentation using k-means and hierarchical clustering
project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING
BACHELOR OF TECHNOLOGY
in
A.Y.: 2026-27

by

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DEPARTMENT OF _____

Name of Title: customer segmentation using k-means and hierarchical clustering

1. Abstract:

The rapid growth of the e-commerce industry has resulted in the generation of vast amounts of customer data, making it essential for businesses to understand customer purchasing behavior and preferences. Customer segmentation is a data-driven approach that enables organizations to divide customers into distinct groups with similar characteristics, allowing businesses to develop personalized marketing strategies and improve customer satisfaction. This project, titled "Customer Segmentation Using K-Means and Hierarchical Clustering in E-Commerce," aims to identify meaningful customer segments by applying two popular unsupervised machine learning algorithms: K-Means Clustering and Hierarchical Clustering. The project utilizes customer transaction data containing attributes such as age, annual income, spending score, purchase frequency, and other behavioral features. Before clustering, the dataset undergoes preprocessing steps including data cleaning, handling missing values, feature selection, and normalization to improve clustering performance. The K-Means algorithm groups customers by minimizing the distance between data points and their respective cluster centroids, while Hierarchical Clustering builds a tree-like structure (dendrogram) to represent similarities among customers. The optimal number of clusters is determined using techniques such as the Elbow Method and Silhouette Score to ensure accurate segmentation. The identified customer groups help businesses recognize high-value customers, loyal customers, potential buyers, and low-engagement customers. These insights enable organizations to design targeted promotional campaigns, recommend personalized products, optimize resource allocation, and enhance customer relationship management. A comparative analysis of K-Means and Hierarchical Clustering is also performed to evaluate their effectiveness in terms of clustering quality, computational efficiency, and interpretability. The proposed system demonstrates how machine learning techniques can transform raw customer data into actionable business insights. By implementing effective customer segmentation, e-commerce companies can improve customer retention, increase sales, support data-driven decision-making, and gain a competitive advantage in today's digital marketplace. This project highlights the practical significance of clustering algorithms in solving real-world business problems and promoting intelligent marketing strategies.

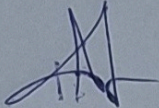
2. INTRODUCTION:

The rapid growth of the e-commerce industry has generated a large amount of customer data, making it important for businesses to understand customer behavior and preferences. Customer segmentation is a data analysis technique that groups customers with similar characteristics, helping businesses improve marketing strategies, enhance customer satisfaction, and increase sales.

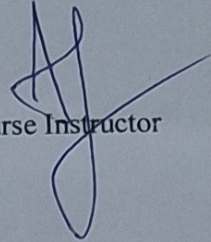
This project, "Customer Segmentation Using K-Means and Hierarchical Clustering in E-Commerce," uses machine learning algorithms to classify customers based on factors such as age, annual income, spending score, and purchase behavior. By comparing K-Means and Hierarchical Clustering, the project identifies meaningful customer groups that help businesses make better decisions, provide personalized recommendations, and strengthen customer relationship management.

3. SOFTWARE REQUIREMENTS:

The proposed system is developed using Python 3.x as the programming language due to its simplicity and extensive support for machine learning and data analysis. The project can be implemented using development environments such as Jupyter Notebook, Visual Studio Code (VS Code), or PyCharm. The dataset used is an E-Commerce Customer Dataset in CSV format. Essential Python libraries include Pandas for data preprocessing, NumPy for numerical computations, Matplotlib and Seaborn for data visualization, Scikit-learn for implementing K-Means and Hierarchical Clustering algorithms, and SciPy for hierarchical clustering analysis. The project can be executed on Windows, Linux, or macOS operating systems.



Signature of the Guide



Course Instructor